

Socioeconomic status related differences in resting state EEG activity correspond to differences in vocabulary and working memory in grade school

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ABSTRACT

There is increasing evidence that children from low income homes exhibit differences in brain, language and cognitive development. To better understand these differences and how they relate to one another, we compared the resting state EEG of forty-five 8–15-year-olds from low-income homes and 45 age and sex matched children from higher income homes who completed a battery of language and cognitive assessments. Children from low income homes performed worse on language tasks and exhibited differences in resting state EEG including more theta and less alpha power. To investigate the relationship between language, cognition and resting state EEG we performed multiple regression analyses. Better working memory was related to lower resting state theta power. There was an interaction between alpha and vocabulary, such that only for children from low income homes, greater resting state alpha was related to higher vocabulary scores. Both relationships held when controlling for resting state changes in the other frequency band, indicating they are somewhat independent effects. These findings shed new light on SES, neural development and cognitive and language outcomes in school-aged children.

1. Introduction

A childhood in a low socioeconomic status (SES) home is associated with exposure to multiple environmental risk factors including poor nutrition, poor prenatal care, increased exposure to violence, lower quantity and quality of child-directed language, higher familial stress and limited structure in daily routines (Evans & English, 2002; Hoff, 2013; Hart & Risley, 1995). Over time, children who are constantly exposed to these environmental factors are more likely to struggle academically with the areas of language and executive functions exhibiting the greatest deficits (Farah, Noble, & Hurt, 2005; Farah et al., 2006; Noble, McCandliss, & Farah, 2007; Noble, Norman, & Farah, 2005). There is increasing evidence that children from low income environments show a difference in neural structure and function (Mackey et al., 2015; Noble, Houston, Kan, & Sowell, 2012; Ursache, Noble, & Blair, 2015; Hoff, 2013). This brain related research often relies on MRI technology. Similar to fMRI, resting state EEG provides a window into brain development by uncovering changes in functional networks, however resting state EEG captures changes in cortical

rhythms with a sub-second temporal resolution that is not accessible using MRI (Bressler, 1995; Bressler & Tognoli, 2006; Britz, Van De Ville, & Michel, 2010). There is increasing information about how these cortical rhythms correspond to important aspects of cognition such as executive control and memory (Barry, Clarke, & Johnstone, 2003; Klimesch, 1999; Klimesch, Sauseng, & Hanslmayr, 2007; Perone, Palanisamy, & Carlson, 2018). Further, EEG systems can be portable allowing for data acquisition in a wide range of low income environments, which could allow researchers to greatly expand the ways by which we define and study the influence of poverty on brain development. Here we use resting state EEG to study differences in neural development between low and higher SES grade school aged children and how such brain differences relate to differences in cognitive and linguistic processes. These ages between middle childhood and adolescence cover a period of substantial language growth and brain development (Giedd et al., 1999; Gogtay et al., 2004; Raznahan et al., 2011), during which variability in brain development can mediate the relationship between SES and academic outcomes (Mackey et al., 2015).

Resting state EEG provides a measure of cortical activity of neural

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oscillations across designated frequency bands (e.g. theta, alpha, beta, gamma) over the scalp. This activity corresponds to synchronized fluctuations of large-scale neural activity. Identifying fast fluctuations of local and global neuronal oscillations at rest allows for new insights into brain development that cannot be obtained with MRI. In typical development, resting state EEG reveals developmental decreases in low-frequency bands (e.g. theta) and increases in higher frequency bands (e.g. alpha, beta, gamma; Marshall, Fox, & the BEIP Core Group, 2004; Brito, Fifer, Myers, Elliott, & Noble, 2016). Progress is being made in linking these resting state EEG changes to known structural changes in the brain. For example, between middle childhood and adulthood though, the age-related decline in low-frequency EEG power (delta and theta) corresponds to declines in grey matter associated with synaptic pruning (Whitford et al., 2007). Importantly, due to its stability on within single subjects, resting state EEG hold particular promise for identifying individual biological signatures (Smit, Posthuma, Boomsma, & De Geus, 2005; Alahmadi, Evdokimov, Kropotov, Müller, & Jäncke, 2016). In fact, resting state activity accounts for a great deal of the variability in evoked responses to task related stimuli (Arieli, Sterkin, Grinvald, & Aertsen, 1996; Fontanini & Katz, 2008). Thus by studying changes in cortical oscillatory networks, we can learn more about the neural infrastructure underlying the development of cognitive and affective abilities in children (Miskovic et al., 2015).

Resting state EEG can vary greatly between populations. For example, infants who were in institutionalized care in the Bucharest Early Intervention project exhibited greater theta (4–8 Hz) band power and less alpha (9–12 Hz) band power than infants who were never institutionalized (Marshall et al., 2004). In a follow up study at age 12, the children who remained in institutionalized care still exhibited greater theta and lower alpha power compared to children who had never been institutionalized and those who had been placed in foster care at a young age (Vanderwert, Zeanah, Fox, & Nelson, 2016). When studying at-risk populations, somewhat similar trends are observed. For example, children from low income homes in Mexico who were in high-risk environments based on economic, social and cultural factors (e.g., maternal education, income, number of siblings in the home, family addictions) exhibited greater theta and less alpha power than their peers in low-risk households (Otero, 1994; Otero, Pliego-Rivero, Fernandez, & Ricardo, 2003). Although being raised in institutionalized care is considerably different from other at-risk environments, the atypical EEG patterns measured in both groups, greater theta and less alpha, appear to be similar. In both cases, because typical brain development involves decreases in low frequency rhythms, such as theta, and increases in higher frequency rhythms, such as alpha (Anderson & Perone, 2018; Harmony et al., 1988; Uhlhaas, Roux, Rodriguez, Rotarska-Jagiela, & Singer, 2010), these EEG differences in children from at-risk populations are generally interpreted as a developmental lag due to an inability to meet basic physiological needs and differences in social and cognitive engagement (Otero et al., 2003; Vanderwert et al., 2016). In the United States, children from low income homes represent a high-risk population, and income-based differences in EEG power have been reported in infants (Otero et al., 2003; Tomalski et al., 2013). To date, little is known about whether differences in resting state EEG related to SES remain in school-aged children or, importantly, how such differences might relate to cognition and language.

In studies of clinical populations, such as children with ADHD, differences in school-aged children's resting state EEG correlate to cognitive function (Lubar, 1991; Monastra et al., 1999). For example, school-aged children (ages 8 years through adolescence) with high levels of resting state theta power are more likely to exhibit learning and attentional disorders (Barry et al., 2003; Clarke, Barry, McCarthy, & Selikowitz, 2002). In typically developing populations, infants resting state EEG correlates to cognitive and language outcomes at ages 4 and 5 years (Gou, Choudhury, & Benasich, 2011). In 3- to 9-year-olds there is a relationship between the resting state theta/beta ratio and executive functions (Perone et al., 2018), however these studies did not

address SES.

Thus, to date, there have been two distinct lines of research. One addressing the relationship between childhood environments and brain development using resting state EEG and the other focused on the relationship between resting state EEG and language and cognition. Our goal is to bridge these areas to understand the relationship between a low SES environment, brain development and cognitive and linguistic outcomes. A handful of studies using MRI have taken a similar approach (Noble et al., 2015; Mackey et al., 2015; Romeo et al., 2018). For example, Noble et al. (2015) found that family income was positively associated with cortical surface area, and that surface area partially mediated the link between income and measures of executive function, including working memory. Mackey et al. (2015) failed to replicate differences in cortical surface area; however, they reported that low income children had less cortical thickness than their higher income peers. Importantly, thicker cortex, especially over temporal and occipital lobes mediated the relationship between SES and statewide academic test performance. Romeo et al. (2018) examined SES related differences in the relationship between 4- to 6-year-old children's brain function and the number of conversational interactions they had with adults. Children who experienced more conversational turns exhibited greater activation in Broca's area during a language task regardless of SES. However, conversational turns and Broca's area activation jointly mediated the relationship between SES and children's language abilities. Taken together, these findings indicate brain structure and function may account for a portion of the relationship between SES and language outcomes. Further, these findings exemplify the importance of investigating the complex relationship between brain development, SES, and language/cognitive outcomes.

Here we apply a similar approach using resting state EEG. Our goal is to identify differences in resting state EEG between low and high SES children and then investigate if and how those brain differences relate to cognitive and linguistic abilities. Our focus is on working memory and vocabulary. These are two areas known to be influenced by childhood SES (Bradley & Corwyn, 2002; Farkas & Beron, 2004; Hackman & Farah, 2009; Raffington, Prindel, & Shing, 2018) and known to influence skills that relate academic success during the school years (Cain, Oakhill, & Lemmon, 2004; Maguire et al., 2018; McCardle, Scarborough, & Catts, 2001).

2. Methods

2.1. Participants

Participants included 45 children from low-income homes and 45 age- and gender-matched children from higher-income homes. We used income as our measure of SES to identify potential differences in resting state EEG. This is because income best represents the material resources available to the child (Duncan & Magnuson, 2005; Noble et al., 2015). Following previous research (Mackey et al., 2015) income status was determined by eligibility for free and reduced lunch. Measures of free and reduced lunch are national benchmarks for low income that are based on family income and size. Eligibility is determined by multiplying the federal income poverty guidelines by 1.3 (free lunch) and 1.85 (reduced lunch; Federal Register, Child Nutrition Program Integrity, 2016). Five low-income children qualified for only reduced lunch, while the remaining forty qualified for free lunch. Participants were all between the ages of 8–15 years ($M = 10.9$, $SD = 2.14$) and were 60% female. All were fluent English speakers in an English-only classroom. Thirty-four participants were Spanish-English bilingual (24 of whom were low-income) based on parental report. Because all participants' primary academic language was English, vocabulary assessment was in English. We included maternal education in our regression analyses to clarify the relationships between income-related resting state EEG differences, maternal education, working memory, and vocabulary. Maternal education was based on self-reports and were then

Table 1

Demographic variables and performance on behavioral assessments between low- and high-income groups.

Measure	No free and reduced lunch <i>M</i> (<i>SD</i>) [Range]	Free and reduced lunch <i>M</i> (<i>SD</i>) [Range]	Significance test
Age	10.98 (2.16)	11.00 (2.15)	$t(88) = -0.05, p = .96$
Number with Mothers who did not complete High School	1 (2.2%)	15 (33.3%)	$t(88) = 7.81, p < .001$
Number of children who are Bilingual	10 (22.2%)	24 (53.3%)	$t(88) = -3.18, p < .005$
Ethnicity (Number of children who are Hispanic/Latino)	8 (17.8%)	36 (80%)	$t(88) = 7.67, p < .001$
Race (Number of children who are Caucasian)	34 (75.6%)	36 (80%)	$t(88) = 1.80, p = .08$
Vocabulary	116.11 (17.02) [87–160]	99.2 (20.74) [62–186]	$t(88) = 4.23, p < .001$
Working Memory ⁺	8.39 (2.78) [4–14]	7.92 (1.78) [4–14]	$F(1,66) = 1.96, p = .17$
Home Environment (CHAOS)	42.9% (11.5%) [18.0–71.67%]	47.68% (9.18%) [18.0–71.67%]	$t(87) = -2.16, p < .05$

Table 2

Correlation results between demographic variables, language and cognitive outcomes, and resting state amplitude when controlling for age, ethnicity, race and bilingualism.

Variables	1.	2.	3.	4.	5.	6.
1. Maternal Education						
2. CHAOS	−0.10					
3. Vocabulary	0.48***	−0.11				
4. Working Memory	0.08	0.09	0.24			
5. Alpha Amplitude	0.10	−0.06	0.20	0.04		
6. Theta Amplitude	−0.05	0.04	−0.07	0.32*	−0.15	

 $p < .05^*$. $p < .01^{**}$. $p < .001^{***}$.

grouped into 5 categories: Did not complete High School (17.8%), received a High School Degree (18.9%), attended Partial College (25.6%), received a College Degree (26.7%), or received a Graduate Degree (11.1%). See Table 1 for more specific demographic information (see Table 2).

2.2. Stimuli and procedure

All participants were accompanied by a parent or guardian and tested individually in the lab when informed consent and assent were obtained in accordance with the University of Texas at Dallas IRB guidelines. For the resting state EEG recordings, participants sat in a chair 1 m from a computer monitor. Participants' EEG was recorded using a Neuroscan 64-electrode Quickcap with CURRY Neuroimaging software and a Neuroscan Synamps2 amplifier. Impedances were below 10 kOhms before starting acquisition and were monitored throughout the study. Participants were instructed to alternate between having their eyes open and closed (resting state) for one minute each, for a total of 8 min. During eyes open trials, a fixation cross appeared on the screen to guide the participant's attention. During eyes closed trials, the screen remained black as participants' eyes were to be closed. An experimenter sat in the EEG booth to monitor children throughout the course of the task. The eyes closed condition is traditionally used because it minimizes input and reduces task demands (for review, see Van Diessen et al., 2015; Anderson & Perone, 2018). As a result, only trials in which the participant's eyes were closed were included in the analysis. All participants completed the English version of the Peabody Picture Vocabulary Task 4th Edition (PPVT-4; Dunn & Dunn, 2007) and the Digit Span (Blackburn & Benton, 1957). Parents completed a questionnaire about their family income and education, child's language history and the Confusion, Hubbub and Order Scale (CHAOS; Matheny, Wachs, Ludwig, & Phillips, 1995) to measure the degree of confusion and disorganization in the child's home environment. Table 1 includes group means, range, and standard deviations on these

behavioral measures.

2.3. EEG pre-processing

Continuous EEG was collected at a sampling rate of 1000 Hz with CURRY Neuroimaging Suite software and analyzed within the EEGLab toolbox of Matlab. All continuous data was high-pass filtered using a Hamming-windowed sinc FIR filter at 0.1 Hz, with a cut-off frequency (6 dB point) of 80 Hz and resampled down to 500 Hz. Following standards in the field (Benasich, Gou, Choudhury, & Harris, 2008; Zhang et al., 2018; Perone et al., 2018), bad electrodes and areas of muscle activity/artifacts were removed from the continuous EEG data by using the clean raw data plug-in within EEGLab. Data was then decomposed using an Independent Components Analysis (ICA; Delorme & Makeig, 2004; Delorme, Makeig, & Sejnowski, 2001). ICA identifies maximally temporally independent signals available in the continuous EEG data. These signals may represent synchronous or partially synchronous activity within one (or possibly more) cortical areas, otherwise they will identify activity from non-cortical sources (e.g., muscle activity, eye-blink artifacts, pulse, etc.). Once the data was decomposed by ICA, these non-cortical components were automatically identified and removed using the multiple artifact rejection algorithm (MARA) plug-in (Winkler, Haufe, & Tangermann, 2011; Winkler et al., 2014). Since the data was collected from a 64-electrode cap, the data was re-referenced using the average across the entire scalp. For each participant there were 4, 1-minute long eyes closed trials. These trials were epoched into smaller, 3000 msec bins. A spherical interpolation of missing channels was conducted on the epoched data.

2.4. Time frequency analysis

The EEGLab toolbox of Matlab was used to conduct time frequency analysis of the EEG. A Fast Fourier Transformation (FFT) over 1 sec Hanning windows with 50% overlap was applied to all data channels between 3 and 30 Hz. This analysis was averaged across all resting state trials (Delorme & Makeig, 2004). To determine statistical significance, while controlling for multiple comparisons, we conducted a monte-carlo cluster-corrected permutation analysis using the FieldTrip toolbox within EEGLab (Oostenveld, Fries, Maris, & Schoffelen, 2011). For this permutation, trials (3 s epochs) are randomly assigned to one of the SES groups and a test statistic (t -test) is computed on a channel by channel basis within a given frequency (e.g., alpha, theta). For each permutation, neighboring electrodes, as defined by the triangulation method, were compared using a non-parametric t -test, and significant t -statistics ($p < 0.05$) were identified. For each permutation, the sum of the neighbor's t -statistics were stored. This was repeated across 1000 permutations, and only clusters that were significant at an alpha of 0.05 were identified. This statistical measure is well-designed within EEGLab Fieldtrip and, therefore, has been calculated specifically to deal with

the type of multidimensional data found in EEG studies.

Developmental studies have reported either absolute or relative EEG power, with both measures having advantages and disadvantages (Marshall, Bar-Haim, & Fox, 2002). Here, we investigated changes in absolute power, similar to work in early and middle childhood (see Takano & Ogawa, 1998; Gou et al., 2011; Gasser, Verleger, Bächer, & Sroka, 1988; Harmony, Marosi, de León, Becker, & Fernández, 1990); however, we included age as a variable of interest in our hierarchical regressions. If absolute power was influenced by factors such as changes in cranial structure, an age effect would be expected for EEG power, and this was not found (Clarke et al., 2002).

For the current analysis we focused on two frequency bands of interest based on previous research: theta (4–8 Hz) and alpha (9–12 Hz). These frequency bands were selected based on their prominence in reports of SES and resting state EEG research in school-aged children (Harmony et al., 1990; Otero et al., 2003; Vanderwert et al., 2016), as well as their relationship to cognitive outcomes (Barry et al., 2003; Klimesch, 1999; Klimesch et al., 2007). Significance was determined by using the previously mentioned cluster-based permutation analysis at an alpha level of 0.05.

3. Results

3.1. EEG results

The monte-carlo cluster correction permutation analysis revealed a significantly less alpha power during resting state for children from low-income homes compared to higher-income peers, $t(44) = 2.45$, $p = .02$ (Fig. 1). The areas of significance identified by the analysis, which are found over left frontal and central regions, all exhibit the same pattern of greater alpha synchrony in the children from low-income homes than those from higher income homes. This permutation analysis also identified a significant difference in the theta band during resting state, driven by greater power increases at right parietal electrodes for low-income children compared to higher-income children, $t(44) = -2.80$, $p = .007$ (Fig. 2).

3.2. Correlation results

The next goal of this study was to investigate the relationship between SES, measures of language and cognitive performance, and resting state EEG. In particular, we were interested in working memory and vocabulary, two areas in which children from low income homes often perform more poorly than their higher income peers (Hackman & Farah, 2009). Because a large portion of the children in the study were bilingual which can influence vocabulary and working memory, we compared the two groups on these factors. We found no significant differences between bilingual and monolingual children related to working memory, but monolingual children performed significantly better on the PPVT ($M = 113.27$, $SD = 20.66$) than bilingual children

($M = 98.41$, $SD = 17.39$), $t(88) = 3.51$, $p = 0.01$. As a result, bilingualism was included in the correlation and multiple regression analyses as a covariate. Using a partial correlation, controlling for age, ethnicity, race and bilingualism, we found a significant relationship between maternal education and vocabulary ($r = 0.48$, $p < .001$) and theta power and working memory ($r = 0.32$, $p = .01$).

3.3. Regression results

Vocabulary and working memory are two areas of cognition that are known to be significantly impacted by a childhood in a low SES home. With the goal of better understanding how resting state data and factors related to income relate to these abilities, we conducted hierarchical multiple regression analyses with vocabulary and working memory as the outcome variables. To determine whether cortical activity during resting state relates to these outcome variables, the mean power within each frequency band for each individual at the electrodes identified as significant in the previous analysis.

3.3.1. Vocabulary regression

One hierarchical multiple regression analysis was conducted with vocabulary, based on the children's PPVT-4 standard scores, as the dependent variable. There were three stages of this model. At stage one, resting state alpha, theta, income, bilingualism, age, ethnicity, race, maternal education, and home environment (CHAOS) were all entered as independent variables. Stage two introduced the two-way interaction terms between resting state (alpha/theta) and income, age and income, and resting state (alpha/theta) and age. The final stage included the omnibus three-way interaction terms between age, income, and resting state (alpha/theta). We included an interaction term between income, age, and resting state EEG as SES differences observed in resting state EEG may be similar to age-related differences in resting state EEG, thus including the interaction terms can better clarify the relationship between these variables of interest.

The hierarchical regression revealed that our predictor variables at stage one explained 32.5% of the variance ($R^2 = 0.33$, $F(9,79) = 4.22$, $p < .001$). Maternal education significantly predicted vocabulary ($b = 4.93$, $p = .03$). Introducing the interaction terms at the second stage of the model explained an additional 5% of the variance ($R^2 = 0.05$, $F(5,74) = 1.28$, $p = .28$). While this did not account for a significant proportion of the variance, the interaction term between income and resting state alpha significantly predicted vocabulary scores ($b = -4.80$, $p = .03$). Thus, income was a significant moderator of the relationship between resting state alpha and vocabulary.

To probe the statistically significant two-way interaction in our multiple regression, follow up simple slope analyses were conducted (Aiken, West, & Reno, 1991; Cohen, Cohen, West, & Aiken, 2003). Models were run to determine the effect of resting state alpha power on vocabulary at high and low levels of income (i.e. 1 SD above and below the mean, respectively). This analysis resulted in a positive relationship

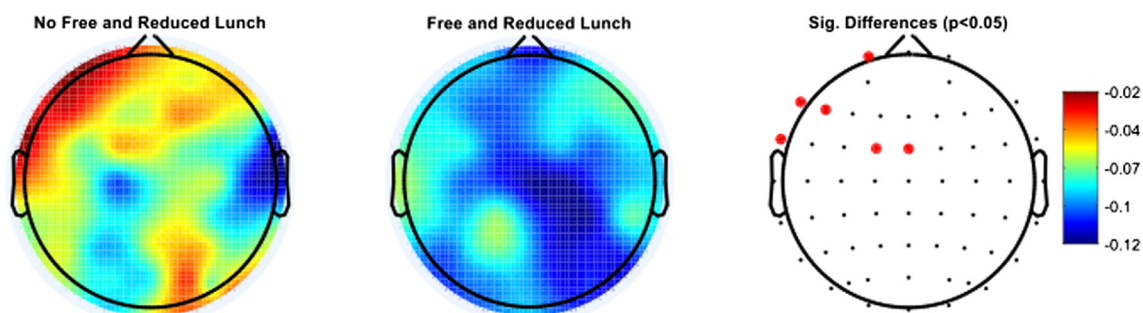


Fig. 1. Resting state alpha differences between groups. Increases (red) in alpha (9–12 Hz) during resting state were significantly greater for higher-income children than lower income children. Significant differences between groups, as identified by the permutation analyses, are denoted by the red dots on the far right scalpmap. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

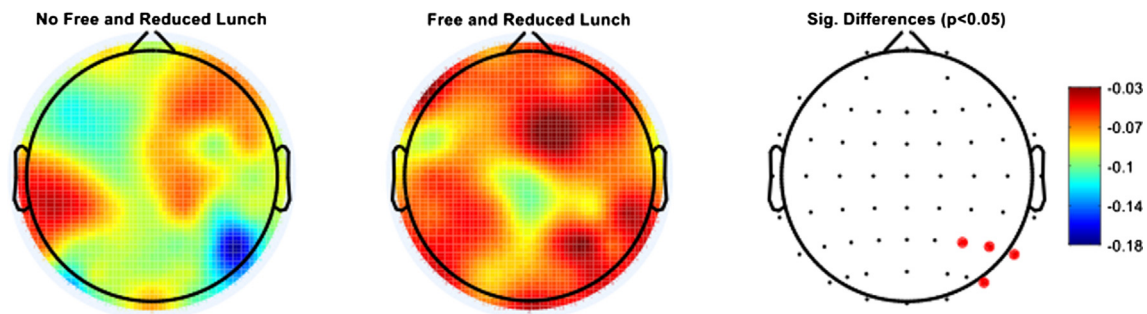


Fig. 2. Resting state theta differences between groups. Increases (red) in theta (4–8 Hz) during resting state were significantly greater for low-income children than higher income children. Significant differences between groups, as identified by the permutation analyses, are denoted by the red dots on the far right scalpmap. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3

Hierarchical multiple regression analysis with vocabulary as the outcome variable.

Variables	β	t	R	R^2	ΔR^2
Step 1			0.57	0.33	0.33
Age	0.01	0.15			
Ethnicity	−0.06	−0.41			
Race	−0.06	−0.58			
Bilingualism	−0.14	−1.04			
Maternal Education	0.31	2.24*			
Income	0.18	1.30			
CHAOS	−0.03	−0.319			
Alpha Amplitude	−0.16	−1.72			
Theta Amplitude	0.15	1.60			
Step 2			0.62	0.38	0.05
Alpha × Income Interaction	−0.23	−2.25*			
Theta × Income Interaction	−0.06	−0.55	–		
Age × Income Interaction	0.09	0.83			
Age × Alpha Interaction	−0.07	−0.57			
Age × Theta Interaction	0.07	0.73			
Step 3			0.62	0.38	0.001
Age × Income Interaction	0.09	0.82			
Age × Alpha Interaction	−0.07	−0.58			
Age × Theta Interaction	0.08	0.76			
Age × Alpha × Income Interaction	0.03	0.23			
Age × Theta × Income Interaction	0.04	0.34			

Note: Income and age were centered at their means.

** $p < .01$, *** $p < .001$.

* $p < .05$.

between alpha amplitude and vocabulary for low-income children ($b = 8.71p = .05$), but not high-income children ($b = 0.19, p = .95$). This effect is independent of the influence of maternal education on vocabulary, which was also a significant predictor of vocabulary ($b = 4.59, p = .04$). The third stage of the model only accounted for an additional 0.1% of the variability in vocabulary ($R^2 = 0.38, F(2,72) = 0.08, p = .93$). The interaction between resting state alpha and income held, as did the main effect of maternal education on vocabulary. Table 3 is a summary of the regression and includes all variables predicting vocabulary scores.

3.3.2. Working memory regression

The second analysis evaluated working memory, based on the children's digit span, as the dependent variable. A hierarchical multiple regression with three stages was conducted with the same independent variables at each stage as in the previous model.

The hierarchical regression revealed that our predictor variables at stage one explained 32.9% of the variance ($R^2 = 0.33, F(9,59) = 3.22, p = .003$). Age significantly predicted working memory ($b = 0.43, p < .001$), which was expected as the digit span does not use scaled scores that are age-normed. Resting state theta also

significantly predicted working memory ($b = 0.17, p = .02$). Thus, changes in resting state theta predicted working memory across all individuals. Introducing the interaction terms in the second stage of the model explained an additional 8.0% of the variance ($R^2 = 0.41, F(5,54) = 1.47, p = .22$). While this did not account for a significant proportion of the variance, resting state theta significantly accounted for variability in working memory performance across all individuals ($b = 0.19, p = .01$). This effect was independent of the influence of age, which was also a significant predictor of working memory ($b = 0.39, p = .002$). In the third stage of the model when the three-way interaction terms were added to the model, only an additional 0.1% of the variability in working memory was accounted for ($R^2 = 0.41, F(2,52) = 0.06, p = .94$). The relationships that were significant at the second stage remained significant. Table 4 is a summary of the regression and includes all variables predicting working memory scores.

4. Conclusions

In this study we examined differences in resting state EEG between low and higher income school-aged children, and how such brain differences may relate to language and cognition. Our findings reveal that children from low income homes exhibit less alpha power and more

Table 4

Multiple regression analysis with working memory as the outcome variable.

Variables	β	t	R	R^2	ΔR^2
Step 1			0.57	0.33	0.33
Age	0.42	3.69***			
Ethnicity	−0.29	−1.52			
Race	0.01	0.08			
Bilingualism	−0.10	−0.63			
Maternal Education	−0.06	−0.34			
Income	0.36	1.75			
CHAOS	0.08	0.76			
Alpha Amplitude	0.07	0.63			
Theta Amplitude	0.27	2.44*			
Step 2			0.64	0.41	0.08
Alpha × Income Interaction	0.04	0.32			
Theta × Income Interaction	0.23	1.92	–		
Age × Income Interaction	0.15	1.16			
Age × Alpha Interaction	−0.03	−0.23			
Age × Theta Interaction	−0.08	−0.74			
Step 3			0.64	0.41	0.001
Age × Income Interaction	0.14	1.08			
Age × Alpha Interaction	−0.02	−0.13			
Age × Theta Interaction	−0.09	−0.65			
Age × Alpha × Income Interaction	−0.05	−0.34			
Age × Theta × Income Interaction	−0.008	−0.06			

Note: Income and age were centered at their means.

** $p < .01$.

* $p < .05$.

*** $p < .001$.

theta power during resting state than their higher income peers. We also performed regression analyses to clarify whether these neurological differences were associated with age and language/cognitive outcomes. We found two unique brain-to-cognition relationships. First, theta power is correlated to working memory. Second, alpha power is correlated to vocabulary in low income children, but not higher income children. Importantly, this is independent of the relationship between maternal education and vocabulary. These findings shed new light on the relationship between a low income environment and brain, cognitive and language development.

Our primary finding, that children from low income homes exhibit more theta power and less alpha power than their higher income peers, reveals a pattern of between-group resting state EEG differences similar to those reported in children from at-risk populations in Mexico (Otero, 1994; Otero et al., 2003) and previously institutionalized children from Romania (Marshall et al., 2004; Vanderwert et al., 2016). Our findings highlight the consistency with which high risk environments are associated with differences in brain development. Importantly, we cannot identify if the resting state differences we observed occur to the same degree as in previous studies of high-risk populations. We suspect they do not. However, the general pattern is the same. As others have noted, typically developing children exhibit a decrease in low frequency rhythms, such as theta, and increases in higher frequency rhythms, such as alpha, and these patterns become more pronounced with age (Marshall et al., 2004; Brito et al., 2016). Although our data do not answer this directly, based on this observation, our findings support the argument that the SES differences observed in resting state EEG are the result of children from low income homes exhibiting a developmental lag compared to their higher income peers, not a deficient or deviant variation in development (Hoff, 2013). This lag is likely due to increased exposure to high risk factors that are more common in low income homes such as poor prenatal, infant and childhood nutrition, increased environmental and home stress, differences in parental responsiveness and involvement, and less exposure to high quality language and cognitive resources (Evans & English, 2002; Hart & Risley, 1995; Hoff, 2013).

Where our findings are particularly unique is in linking the differences in resting state EEG that we observed based on family income to specific cognitive and linguistic outcomes that are known to be problematic for children from low income homes, specifically working memory and vocabulary. Both theta and alpha were found to correspond to unique cognitive or language outcomes, independent of one another, and independent of maternal education, which is highly correlated to income. Thus, these findings suggest that children from low income homes are not exhibiting global differences in brain development that manifest as differences in both alpha and theta, which then correspond to general poor language and cognitive outcomes. Instead, it appears that different aspects of brain, language and working memory development uniquely relate to a childhood in a low income environment. To some degree this finding supports similar research using MRI which reveals that specific areas of the brain are more vulnerable to the effects of a low income environment than others (Bruto, Piccolo, & Noble, 2017; Raizada, Richards, Meltzoff, & Kuhl, 2008; Noble et al., 2015). However, it expands on that work by extending it to resting state EEG, revealing that SES-related brain differences in resting state theta and alpha are statistically independent from one another.

The most fascinating set of findings pertains to vocabulary and highlights the many paths that lead to differences in language abilities between low and higher income children. The regression identified two significant predictors of vocabulary: maternal education and the interaction term between income and resting state alpha. Maternal education is a well-established predictor of vocabulary size in young children (Hoff, 2003). Here we find that even in children ages 8- to 15-years, an age when children are spending a great deal of time in school, maternal education continues to predict vocabulary.

Our EEG findings reveal that there is a second, independent path

that also relates to vocabulary differences in children from low income homes. Specifically, alpha amplitude predicts vocabulary, but only in children from low income homes. The fact that this relationship is only found in children from low income homes is in line with previous findings about the relationship between SES, brain development and cognitive function. For example, Brito et al. (2017) reported that family income and parental education moderated the association between cortical thickness and cognitive skills, specifically executive functions and language, in such a way that coming from a low SES home appeared to exaggerate the relationship between brain structure and cognitive function in children. A similar process may account for the current relationship between resting state alpha and vocabulary, which was only significant in children from lower income families. Another possibility that may account for the relationship between alpha and vocabulary in lower income, but not higher income children, is the high degree of variability in vocabulary ability within the low income group. The high variability within this group may have allowed for significant relationships between vocabulary and resting state alpha to be uncovered that were not as strong in high income children who all perform relatively well on the task. Why alpha in particular was related to vocabulary, independent of the influence of maternal education, in this age group is unclear. The research related to identifying relationships between resting state EEG and cognition is still quite sparse (Perone et al., 2018). Alpha is most commonly associated with attention/inhibition (Jensen & Mazaheri, 2010; Klimesch et al., 2007; Maguire, Brier, & Ferree, 2010; Meyer, Obleser, & Friederici, 2013). Therefore, it may be that alpha is related to a more general learning mechanism, which results in difficulty acquiring new words independent of the child's environment (Maguire et al., 2018). Future research will need to address this question.

Our finding that differences in theta power are related to working memory is grounded in previous work establishing theta's role in working memory (Klimesch, 1999; Gevins et al., 1998; Raghavachari et al., 2001, 2006; Böcker et al., 2010). What was somewhat surprising is the lack of differences between groups on the digit span task, which was our measure of working memory. Previous reports suggest that SES differences in brain structure (Mackey et al., 2015; Noble et al., 2015) and function (Romeo et al., 2018) can mediate the relationship between SES and cognitive outcomes in children. We believe that a wider range of working memory tasks in the future may allow us to better understand the direct and indirect effects of SES on working memory and the brain's role in that relationship.

The topography of the theta and alpha differences in the current study are quite distinct. In an attempt to link our findings to previous work in the field, we found that past findings are considerably mixed regarding the localization of alpha and theta during resting state. This is likely attributed to substantial variability in task (eyes open versus closed, length of time, etc.). Additionally, many researchers have focused only on a priori predictions related to frontal regions, whereas the current study used data driven approaches to examine localization differences between groups. Anderson and Perone (2018) provide an excellent summary of past resting state EEG studies across the course of development, highlighting localization differences. This is an area where future research is greatly needed.

The evidence presented here does not suggest that SES causes brain differences. We suspect that the group differences we observed are due to children from low income homes being more likely to experience a greater number of risk factors that, over time, correspond to differences in the rate of brain development. Although we did not uncover a three-way interaction between resting state EEG, developmental age, and income, it is possible that our null findings are attributed to a lack of power. Future research should continue to disentangle whether age plays a role in the interaction between income and resting state EEG. Another limitation of this study is that to identify differences in resting state EEG we measured SES using only one, binary categorical measure: qualifying for free and reduced lunch. This is an economic measure that

takes into account important factors such as family income and number of people in the home, but masks over the nuanced complexity of SES. In future work with larger samples, a more detailed investigation into the specific aspects of SES that underlie these resting state differences is necessary.

These findings highlight the utility of using resting state EEG to study the impact of poverty on brain development. Previous research has identified structural differences between low and higher SES children using MRI (Jednoróg et al., 2012; Lawson, Duda, Avants, Wu, & Farah, 2013; Mackey et al., 2015; Noble et al., 2012, 2015; Piccolo et al., 2016). Our findings provide evidence that EEG can be a reliable tool for identifying a relationship between SES, resting state activity, and cognitive outcomes in grade school aged children. However, this line of research is still in its infancy. What is necessary is a thorough understanding of the factors that leads to these differences in neural activity and cognitive abilities. More detailed information about the child's environment such as nutrition, prenatal health, exposure to violence, and school may help tease apart these factors.

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Declaration of Competing Interest

None.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bandc.2019.103619>.

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